Fiscal Year:	FY 2009	**Task Last Updated:**	FY 12/23/2008
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	03/01/2007	**End Date:**	02/28/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Morfill, Gregor (Max Planck Institute for Extraterrestrial Physics)		
Grant/Contract No.:	NNX07AD22G		
Performance Goal No.:			
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This fundamental research is an experimental study of dusty plasma physics. Data are analyzed from a European instrument, PK-3 Plus, presently on orbit on the International Space Station ISS. (The acronym PK refers to "Plasma Crystal," and ISS is the International Space Station.) Ground-based laboratory experiments are carried out to prepare for future microgravity flight experiments, including the planned PK-4 instrument.
Dusty plasma consists of particles of solid matter suspended in ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma.

Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We study: • Structural

Task Description:	features. These include the boundaries of the particle suspension, including a void at the plasma's center and vortices at its edges. • Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, and transport. This dynamical behavior is studied at an atomistic level, using video micrography to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter. Experimental methods include direct imaging of particle suspensions using video micrography and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers. Ground-based laboratory experiments accompany the analysis of data returned from space. The ground-based work serves the dual purpose of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return. The spaceflight microgravity experiments are performed using European instruments on ISS: PK-3 Plus (presently on orbit) and PK-4 (planned). NASA is not expected to provide any flight hardware. Video cameras are used to image not only the particle suspension as in previous experiments, but also spectral emission from the plasma glow. These space-flight experiments on ISS are carried out under the direction of German and Russian scientists. The PI is a team member on both the PK-3 Plus and PK-4 missions, and receives the data after it is returned from the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This report covers the period March 1, 2008 - February 28, 2009. The report was written December 23, 2008, and the progress between that date and February 28, 2009 is projected rather than actual progress, as requested by NASA's ESMD Advanced Capabilities Division. During this period, the PI, Professor Goree, became a team member for two missions on the International Space Station (ISS). The PK-3 Plus mission is primarily a German-Russian collaboration, and the instrument is presently on-orbit on the ISS and returning data. The PK-4 mission is primarily a European Space Agency (ESA) mission, and is now in the planning stages. The acronym PK refers to the German spelling of "Plasma Crystal." Both projects concern dusty plasma, also known as complex plasma, which consists of micron-size particles of solid matter immersed in ionized gas. During this period, Professor Goree received two shipments of data from PK-3 Plus, and his team began data analysis. Data that were returned aboard Soyuz spacecraft were delivered to the German team in Garching Germany, and from there to Professor Goree in Iowa. The data consist of digitally-recorded video accompanied by time series recordings of voltage, temperature, and other housekeeping data. Two scientific topics are being explored at Iowa using PK-3 Plus data. First, waves in a single-layer suspension of microparticles are observed. These waves have a spectrum of frequencies and wavenumbers, corresponding to natural random thermal motion. We are computing the experimentally-observed spectrum, using the PK-3 Plus data, and comparing these to theoretical dispersion relations. The theory is for sound waves in a monolayer suspension of particles that interact through a shielded Coulomb repulsion. Second, the spectral emission of neutral gas atoms in the plasma are being compared with and without dust, using the PK-3 Plus data, to infer the electric field conditions in the plasma. During this period, Professor Goree also carried out ground-based laboratory research to support the flight experiments. Microgravity enables the study of three-dimensional suspensions of microparticles, unlike laboratory conditions where the particles usually sediment to form a thin horizontal layer. To test manipulation schemes that will be useful in future microgravity experiments, including PK-4, we are performing laboratory tests using powerful lasers to manipulate particles. The radiation pressure of these lasers pushes particles, and sets the collective body of microparticles into a liquid-like flow pattern. To simulate microgravity conditions, we apply an upward thermophoretic force, by applying a vertical temperature gradient to the gas in our vacuum chamber. This partially offsets the downward force of gravity, allowing us to form a three-dimensional suspension of limited size in the laboratory. This kind of three-dimensional suspension, called a Yukawa Ball, was recently developed by collaborators in Kiel, Germany. Computer simulations of the experiments are an important part of analyzing experimental data as well as preparing for future experiments. During this period, Professor Goree's team collaborated with a leading simulation group in Budapest Hungary, led by Dr. Zoltan Donko. The simulations tested whether transport coefficients, such as diffusion, viscosity and thermal conductivity, are meaningful for an analog liquid with shielded Coulomb interparticle interactions. During CY 2008, three papers were published in Physical Review Letters, two in Physical Review, and one in Physics of Plasmas.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	Feng Y, Liu B, Goree J. "Rapid heating and cooling in two-dimensional Yukawa systems." Phys Rev E Stat Nonlin Soft Matter Phys. 2008 Aug;78(2 Pt 2):026415. PMID: 18850954 , Aug-2008
Articles in Peer-reviewed Journals	Liu B, Goree J, Feng Y. "Non-Gaussian statistics and superdiffusion in a driven-dissipative dusty plasma." Phys Rev E Stat Nonlin Soft Matter Phys. 2008 Oct;78(4 Pt 2):046403. PMID: 18999539 , Oct-2008
Articles in Peer-reviewed Journals	Donkó Z, Goree J, Hartmann P, Liu B. "Time correlation functions and transport coefficients of two-dimensional Yukawa Liquids." Physical Review E, in press, January 2009. , Jan-2009
Articles in Peer-reviewed Journals	Sheridan TE, Nosenko V, Goree J. "Experimental study of nonlinear solitary waves in two-dimensional dusty plasma." Physics of Plasmas, 2008 Jul;15(7):073703. http://dx.doi.org/10.1063/1.2955476 , Jul-2008
Articles in Peer-reviewed Journals	Nosenko V, Zhdanov S, Ivlev AV, Morfill G, Goree J, Piel A. "Heat transport in a two-dimensional complex (dusty) plasma at melting conditions." Physical Review Letters. 2008 Jan 18;100(2):025003. PMID: 18232879 , Jan-2008
Articles in Peer-reviewed Journals	Liu B, Goree J. "Superdiffusion and non-Gaussian statistics in a driven-dissipative 2D dusty plasma." Physical Review Letters. 2008 Feb 8;100(5):055003. PMID: 18352381 , Feb-2008

Articles in Peer-reviewed Journals	Feng Y, Goree J, Liu B. "Solid superheating observed in two-dimensional strongly coupled dusty plasma." Physical Review Letters. 2008 May 23;100(20):205007. PMID: 18518548 , May-2008